

CBCT Segmentation in Head and Neck ART: A Longitudinal Deep Learning Approach

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Head and Neck Cancer (HNC)



› Progressive anatomical changes:

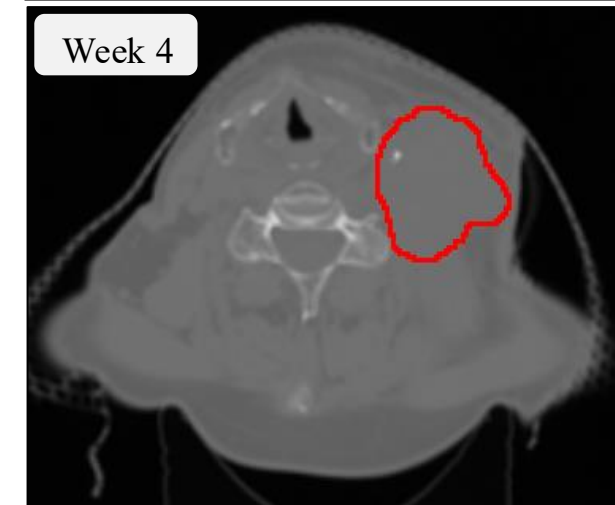
- › Weight loss
- › Tumor and lymph nodes shrink $\sim 3\%$ / day
- › Other OARs (including the parotid glands) also lose a significant part of their volume

› Random anatomical variations:

- › Neck flexion
- › Sinus filling



Potential **discrepancies** between **planned** and **delivered** dose distributions



■ Tumor

Cone-Beam CT (CBCT)



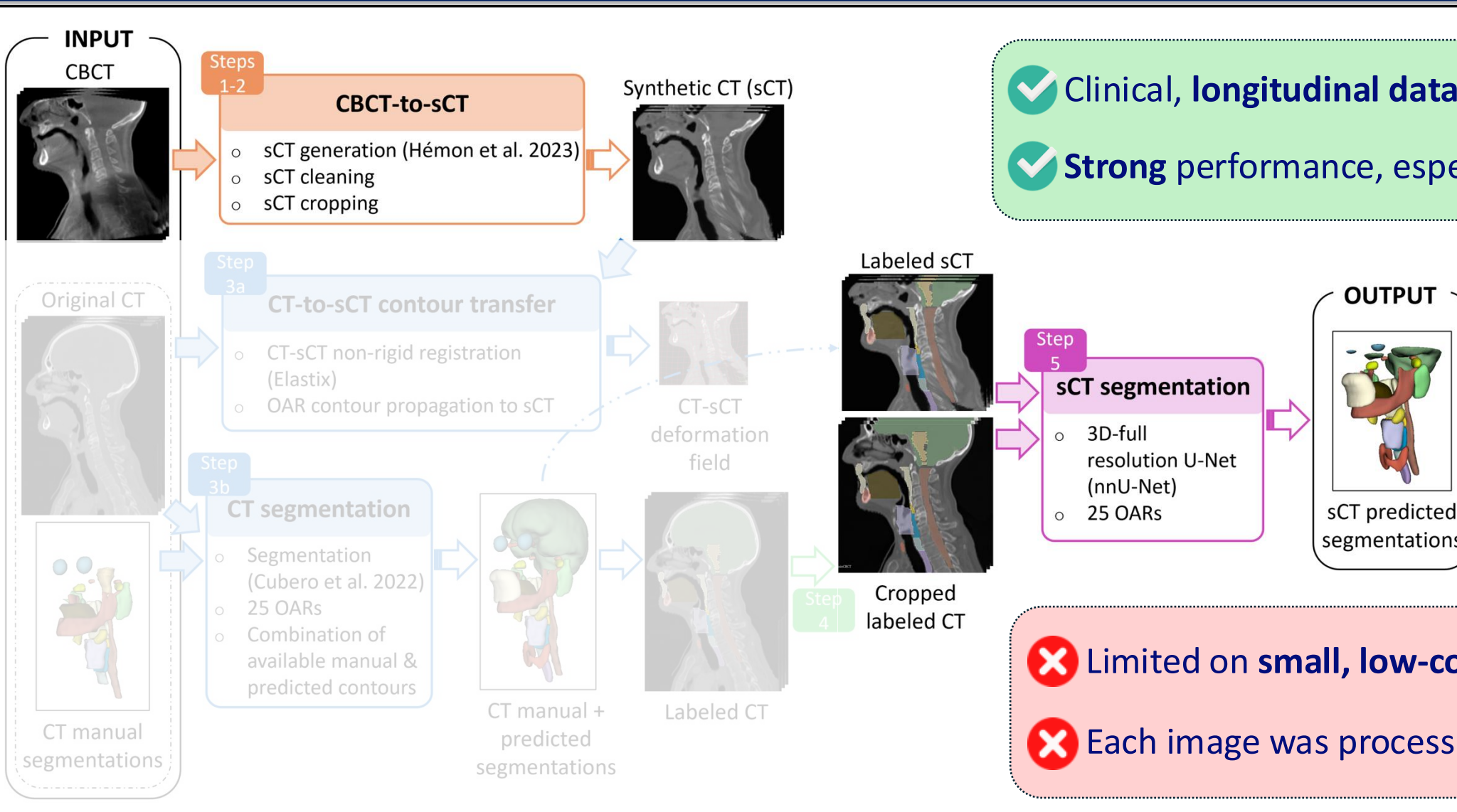
Adaptive Radiotherapy (ART): treatment plan adaptation to account for anatomical changes through repeated imaging.



- **OAR segmentation on longitudinal CBCT** can be a key component for **dose monitoring** and ART
- Challenging due to:
 - × Lower image quality
 - × Limited field-of-view
 - × Artifacts
 - × Hounsfield units inconsistency
 - × Scarcity of annotated CBCT datasets

**Synthetic CT (sCT)
generation**

Motivation · Previous work



- ✓ Clinical, longitudinal data
- ✓ Strong performance, especially on large OARs

- ✗ Limited on small, low-contrast OARs
- ✗ Each image was process independently

Motivation · Longitudinal segmentation



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Original Research Article

Progressive auto-segmentation for cone-beam computed tomography-based online adaptive radiotherapy

Hengrui Zhao, Xiao Liang, Boyu Meng, Michael Dohopolski, Byongsu Choi, Bin Cai, Mu-Han Lin, Ti Bai, Dan Nguyen^{*}, Steve Jiang^{*}

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Most segmentation studies
do not exploit the
longitudinal information

- Progressive **3D LSTM-UNet**
- Recurrent **memory**
- **Retain** and **update** information



One separate **BINARY** model per OAR



Trained on a **SIMULATED** longitudinal **dataset**



1 CBCT SEGMENTATION

- Develop a **multi-OAR longitudinal segmentation** model for CBCT
- Clinical HNC ART cohort

2 DOSIMETRIC ANALYSIS

Assess its potential for:

- **Longitudinal dose monitoring**
- Identification of **patients** who may benefit from **ART**

Database

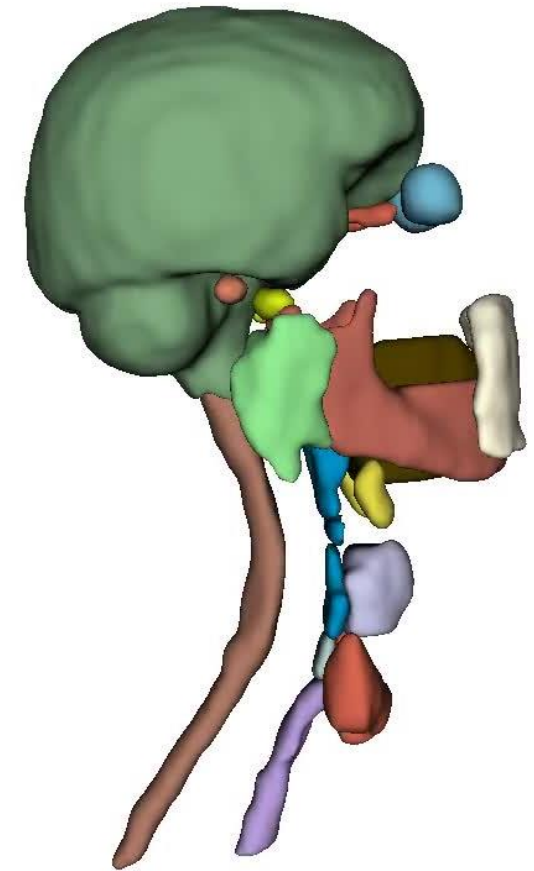
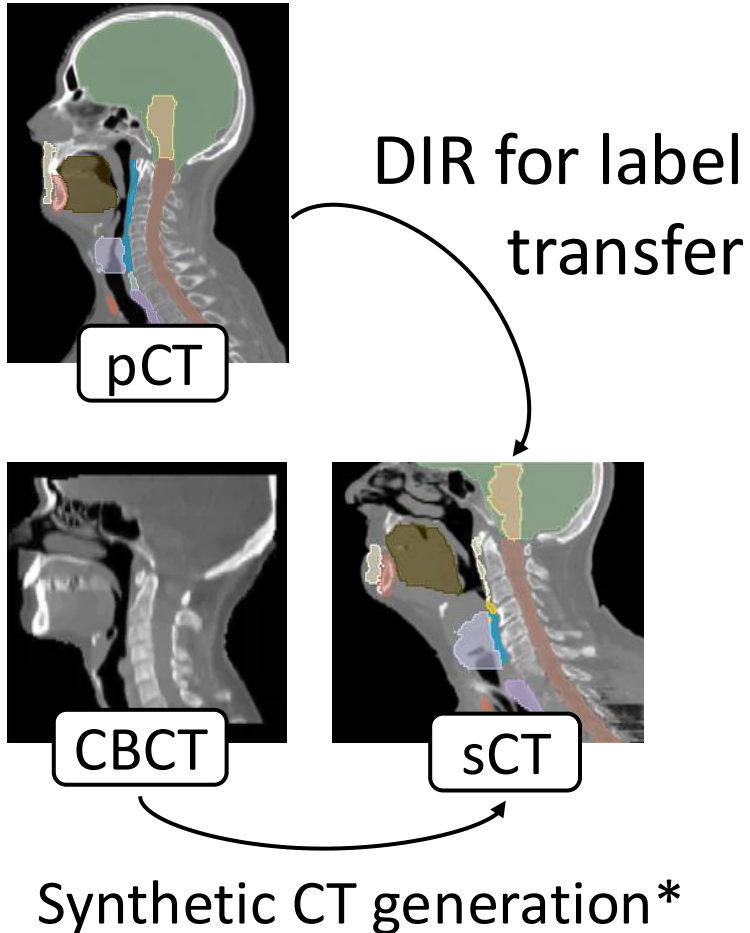


	TRAIN	TEST
# patients	129	26
RT protocol	Standard RT	Weekly replanning (ARTIX trial*)
# CBCT	818	156
Planning	129	26
# CT Replanning	95	77
Total	224	103

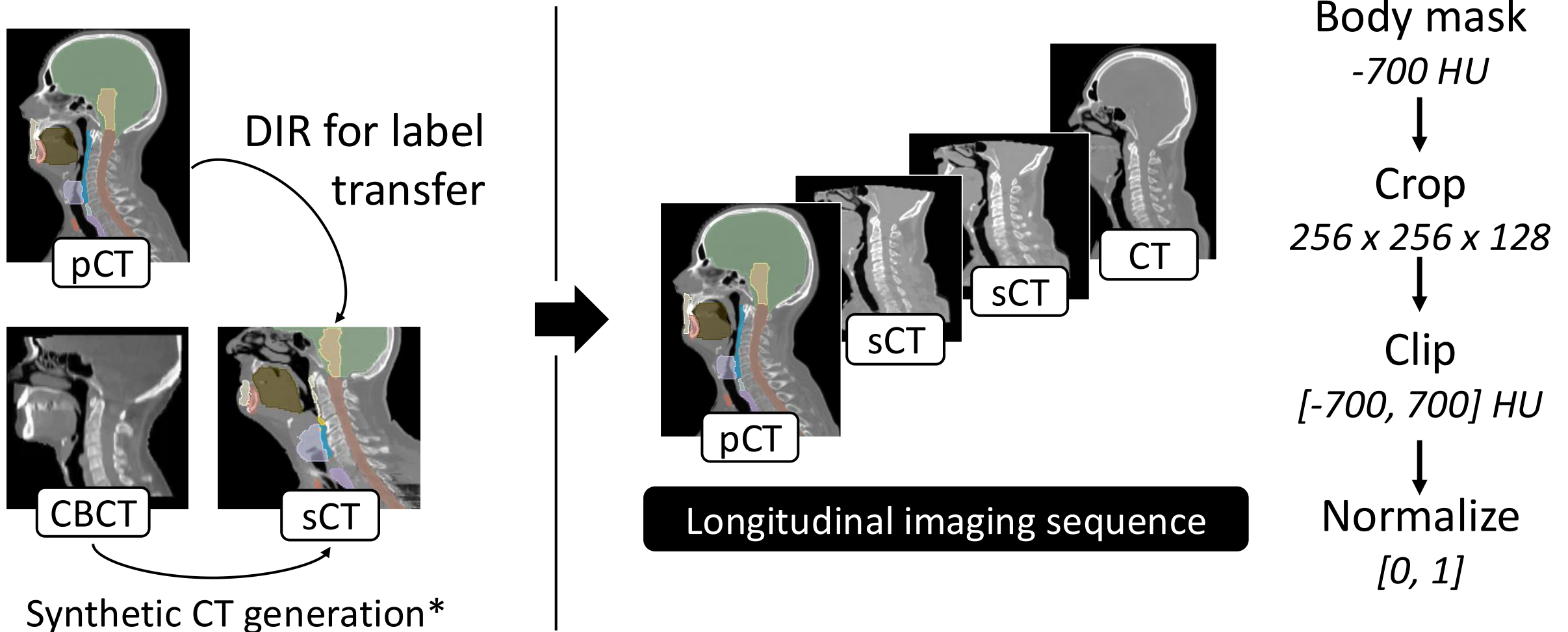
Preprocessing



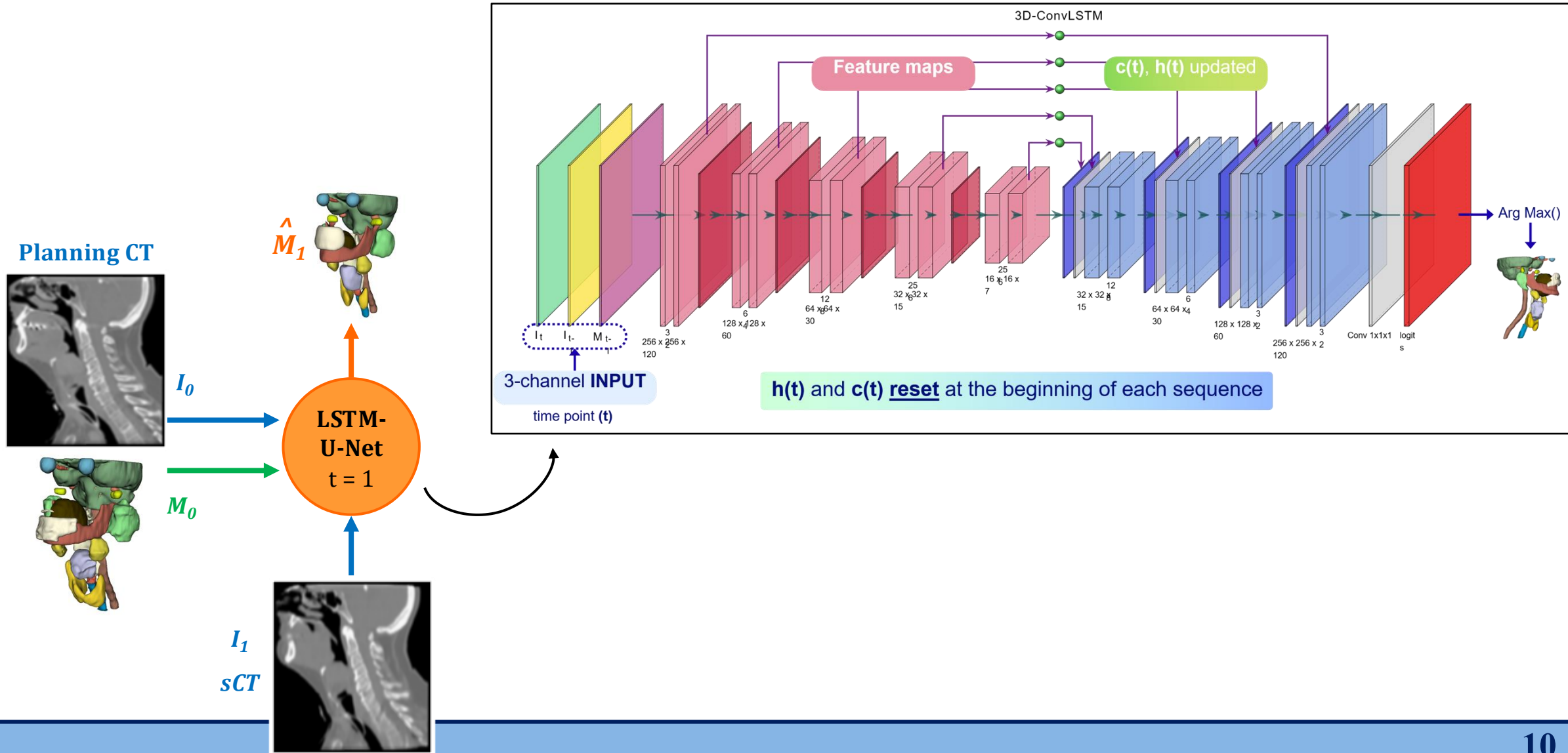
17 HN
OAR classes



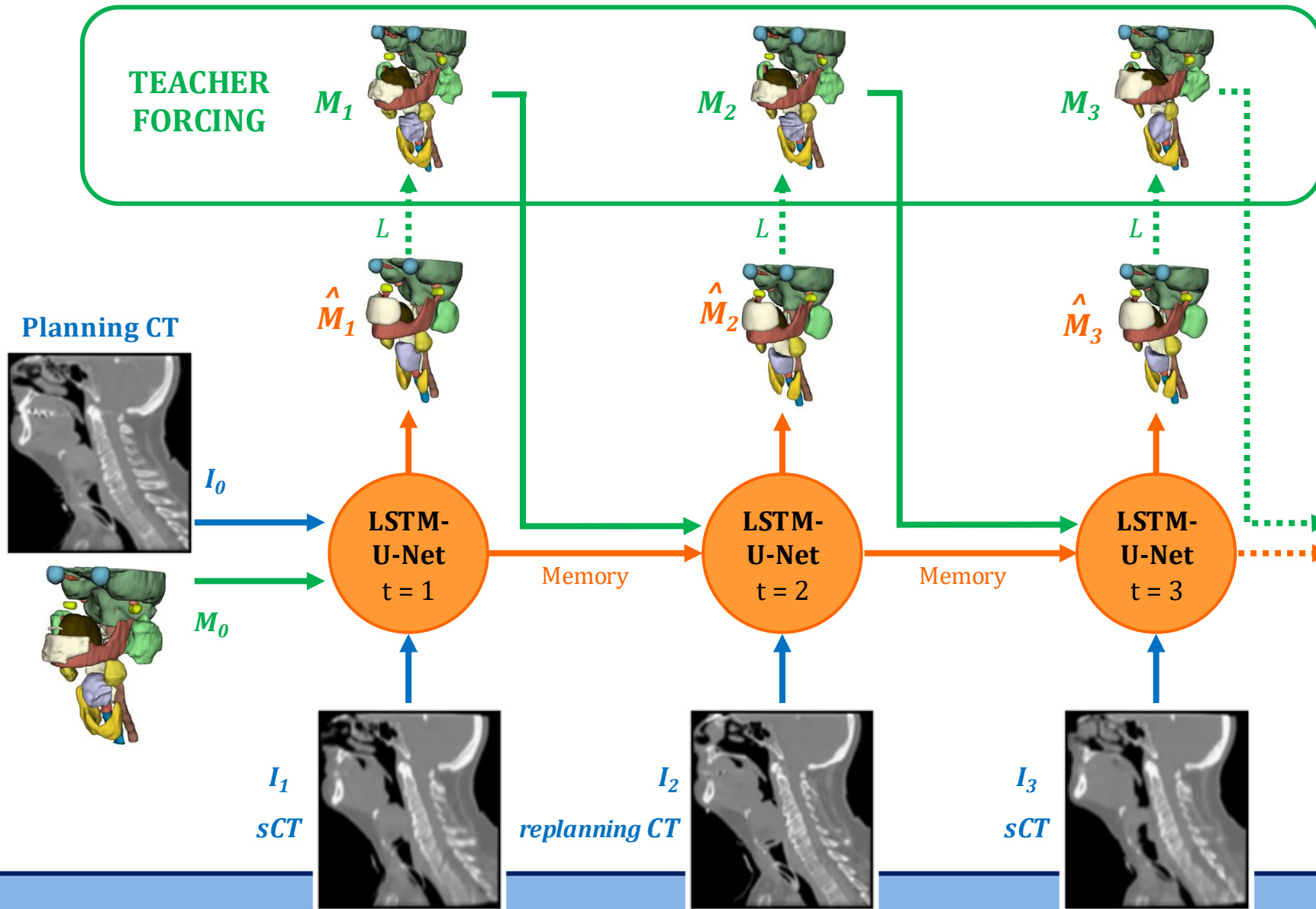
Preprocessing



Longitudinal Model



Longitudinal Model



TRAINING

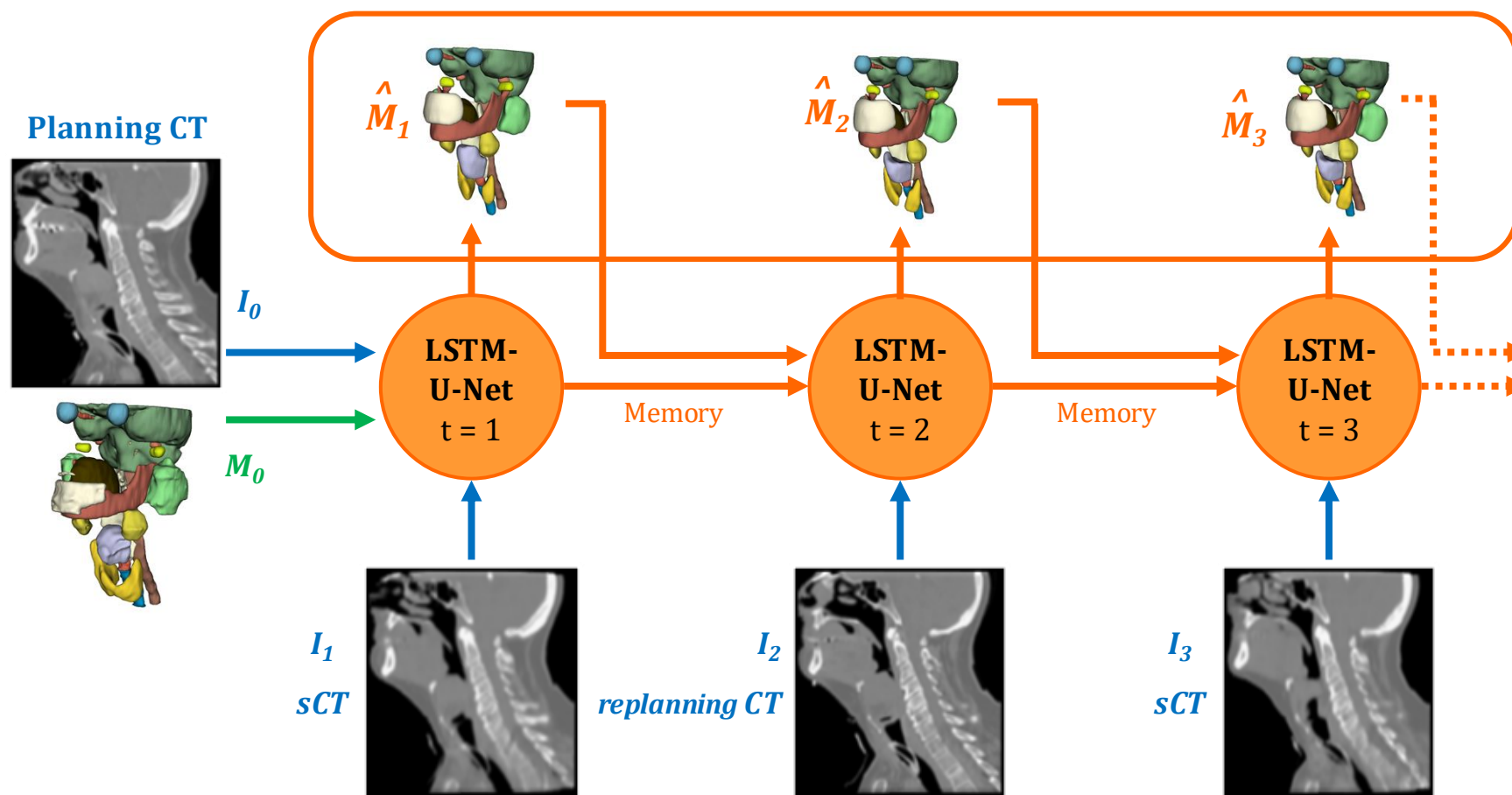
- Dice CE loss
- Previous mask dropout ($p=0.15$)

Inference Settings

After the initial prior, the model uses its own prediction as input for the next scan

$$\hat{M}_{t-1}$$

AUTOREGRESSIVE

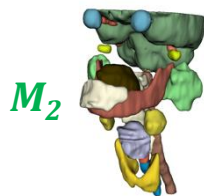


Inference Settings

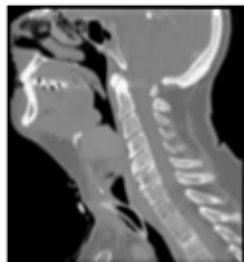
Predictions are propagated autoregressively from sCT, but for replanning CTs, GT masks are used as priors for the next scan.

M_{t-1}

REPLANNING
GT-RESET



Planning CT

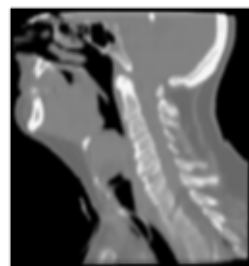


I_0

M_0

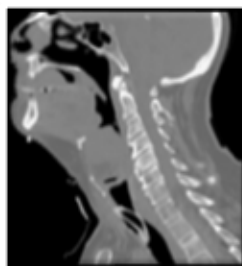


I_1
sCT

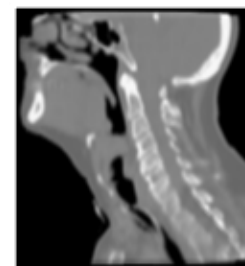


replanning CT

I_2



I_3
sCT



LSTM-
U-Net
 $t = 1$

LSTM-
U-Net
 $t = 2$

LSTM-
U-Net
 $t = 3$

Memory

Memory

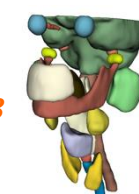
$\hat{M}_1$



$\hat{M}_2$



$\hat{M}_3$



Results

OAR	nnU-Net
Brain	95.6 ± 2.0
Brainstem	81.8 ± 4.6
Spinal canal	85.4 ± 5.8
Eyeballs	82.4 ± 8.2
Optic nerves	66.6 ± 13.6
Inner ears	66.4 ± 12.6
Parotid glands	80.6 ± 4.7
Mandible	88.3 ± 2.1
TMJs	69.8 ± 8.0
Oral cavity	82.8 ± 6.1
Lips	64.4 ± 12.1
SMGs	71.1 ± 7.8
Larynx	64.7 ± 13.3
PCM	65.7 ± 9.6
Thyroid	76.4 ± 10.0
Esophagus	70.5 ± 10.1
Esophageal inlet	44.3 ± 23.3
Mean	73.9 ± 15.6

OAR	nnU-Net	Autoregressive
Brain	95.6 ± 2.0	94.3 ± 4.5*
Brainstem	81.8 ± 4.6	82.2 ± 12.3*
Spinal canal	85.4 ± 5.8	83.9 ± 8.2*
Eyeballs	82.4 ± 8.2	78.0 ± 11.3*
Optic nerves	66.6 ± 13.6	62.4 ± 15.1*
Inner ears	66.4 ± 12.6	70.4 ± 14.8*
Parotid glands	80.6 ± 4.7	76.3 ± 9.3*
Mandible	88.3 ± 2.1	85.4 ± 5.5*
TMJs	69.8 ± 8.0	71.1 ± 8.5*
Oral cavity	82.8 ± 6.1	81.5 ± 5.3
Lips	64.4 ± 12.1	70.9 ± 9.2*
SMGs	71.1 ± 7.8	69.3 ± 9.0
Larynx	64.7 ± 13.3	71.0 ± 13.6*
PCM	65.7 ± 9.6	64.5 ± 12.9
Thyroid	76.4 ± 10.0	71.7 ± 11.5*
Esophagus	70.5 ± 10.1	74.3 ± 8.2*
Esophageal inlet	44.3 ± 23.3	46.4 ± 23.3
Mean	73.9 ± 15.6	73.8 ± 15.6

Results

› **Autoregressive inference competitive** with nnU-Net.

OAR	nnU-Net	Autoregressive	Replan.
Brain	95.6 ± 2.0	94.3 ± 4.5*	94.6 ± 4.6*
Brainstem	81.8 ± 4.6	82.2 ± 12.3*	84.8 ± 12.5*
Spinal canal	85.4 ± 5.8	83.9 ± 8.2*	85.9 ± 8.4*
Eyeballs	82.4 ± 8.2	78.0 ± 11.3*	80.8 ± 11.2
Optic nerves	66.6 ± 13.6	62.4 ± 15.1*	63.8 ± 15.4*
Inner ears	66.4 ± 12.6	70.4 ± 14.8*	73.4 ± 14.6*
Parotid glands	80.6 ± 4.7	76.3 ± 9.3*	81.4 ± 10.0*
Mandible	88.3 ± 2.1	85.4 ± 5.5*	86.8 ± 5.5*
TMJs	69.8 ± 8.0	71.1 ± 8.5*	74.4 ± 8.5*
Oral cavity	82.8 ± 6.1	81.5 ± 5.3	85.7 ± 5.4*
Lips	64.4 ± 12.1	70.9 ± 9.2*	77.0 ± 8.6*
SMGs	71.1 ± 7.8	69.3 ± 9.0	74.0 ± 9.1*
Larynx	64.7 ± 13.3	71.0 ± 13.6*	77.3 ± 12.7*
PCM	65.7 ± 9.6	64.5 ± 12.9	70.0 ± 10.6*
Thyroid	76.4 ± 10.0	71.7 ± 11.5*	75.6 ± 9.9
Esophagus	70.5 ± 10.1	74.3 ± 8.2*	77.5 ± 7.4*
Esophageal inlet	44.3 ± 23.3	46.4 ± 23.3	51.5 ± 23.7*
Mean	73.9 ± 15.6	73.8 ± 15.6	77.4 ± 14.9

Results

- › **Replanning-guided inference** achieved the **highest mean DSC**.
 - › Including ground truth contours for replanning CTs was beneficial.
- › **12/17 OARs significantly improved vs. nnU-Net (*)**.
 - › Largest gains: larynx (+12.6%), lips (+12.6%), esophageal inlet (+7.2%), esophagus (+7.0%).

DSC (%) by OAR

16

(*) significant vs. nnU-Net (Wilcoxon, Holm-corrected)



1 CBCT SEGMENTATION

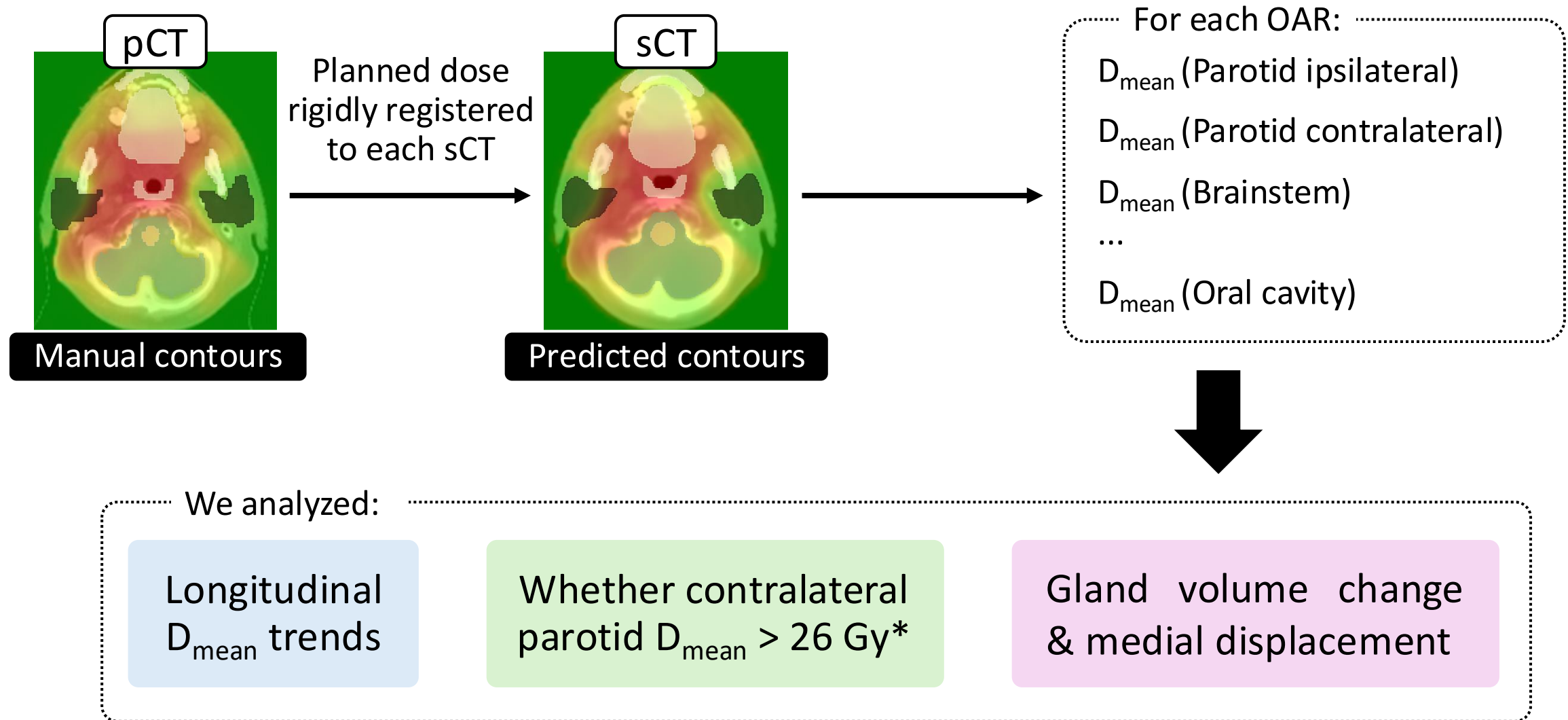
- Develop a **multi-OAR longitudinal segmentation** model for CBCT
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2 DOSIMETRIC ANALYSIS

Assess its potential for:

- **Longitudinal dose monitoring**
- Identification of **patients** who may benefit from **ART**

Dosimetric Analysis



Dosimetric Analysis · Results

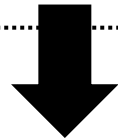
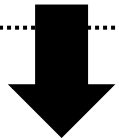


We analyzed:

Longitudinal
 D_{mean} trends

Whether contralateral
parotid $D_{\text{mean}} > 26 \text{ Gy}^*$

Gland volume change
& medial displacement



D_{mean} increased significantly for:

Parotid (ipsi.)	+0.60 Gy/week
Parotid (contra.)	+0.56 Gy/week
Submandibular (ipsi.)	+0.17 Gy/week
Submandibular (contra.)	+0.25 Gy/week



46% of patients (12/26)
exceeded 26 Gy
constraint during
treatment

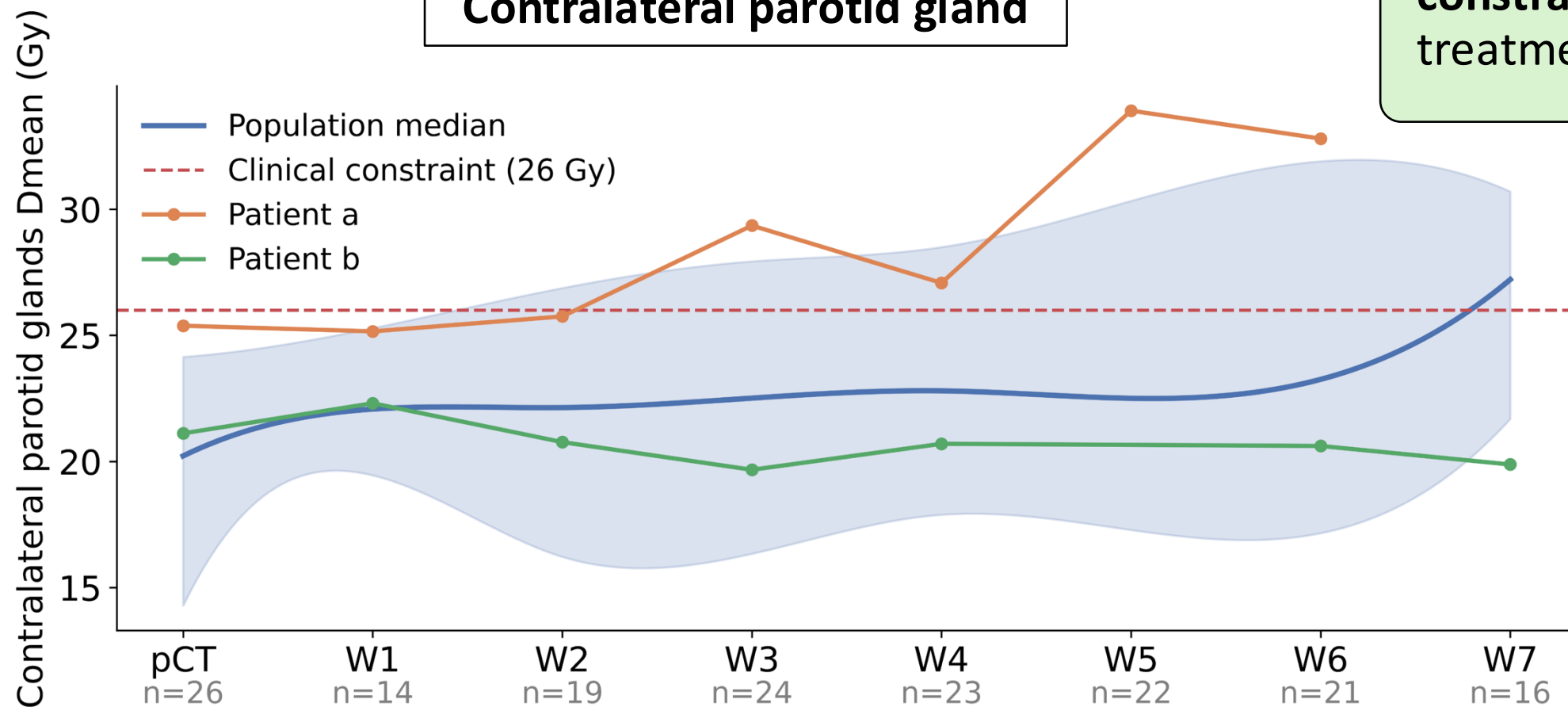


D_{mean} increase
correlated with
medial displacement
(+0.31 mm/week)
with $p = 0.03$

Dosimetric Analysis · Results



Contralateral parotid gland



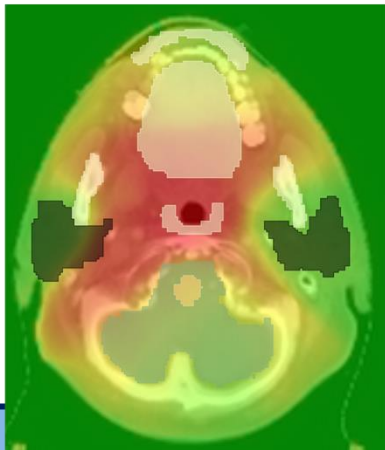
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Dosimetric Analysis · Results



**46% of patients (12/26)
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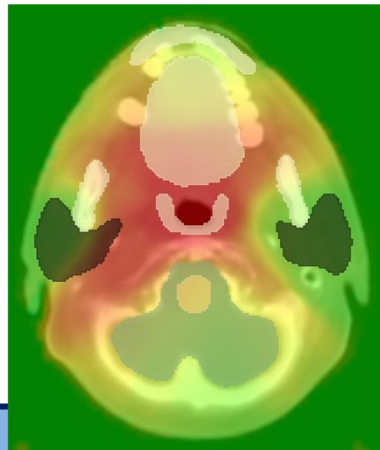
pCT
 $D_{\text{mean}} = 21.11 \text{ Gy}$



Week 1 (CBCT)
 $D_{\text{mean}} = 22.31 \text{ Gy}$



Week 3 (CBCT)
 $D_{\text{mean}} = 21.99 \text{ Gy}$



Week 6 (CBCT)
 $D_{\text{mean}} = 20.61 \text{ Gy}$



Patient (b)

Take home messages



CBCT SEGMENTATION

- **Longitudinal memory-based segmentation on CBCT** extended to 17 OARs.
- **Replanning-guided inference** achieved 77.4% mean DSC, significantly improving 12/17 OARs over nnU-Net.

DOSIMETRIC ANALYSIS

- Longitudinal CBCT segmentation enabled **patient-specific dose monitoring**
- **46% of patients exceeding 26 Gy contralateral parotid constraint during treatment.**

- › No manual sCT annotations
- › Rigid dose propagation

- › Multicenter validation
- › Uncertainty-aware temporal updates to account for autoregressive errors

The framework provides a **foundation for personalized longitudinal monitoring for ART.**

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